

AVI BENJAMIN SHEIN

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EDUCATION

University of California, Irvine, ICS, Irvine, CA September 2026 - Present
PhD in Computer Science

Northwestern University, TGS, Evanston, IL September 2024 - June 2026

Master of Science in Computer Science; GPA: 3.97

Thesis: [Theoretical Bounds and Structure of Byte-pair Encoding](#) (Advisor: Konstantin Makarychev)

Relevant Courses: Approximation Algorithms, Cryptography, Algorithmic Game Theory, Quantum

Tufts University, School of Engineering, Medford, MA May 2024

B.S. in Computer Science, Math Minor; GPA: 3.9 (*summa cum laude*), 6 semester TA (Algorithms, Discrete)

Relevant Courses: Computational Geometry, Graph Theory, Theory of Computation, Abstract Algebra

EXPERIENCE

EndoSec LLC (Security and Cryptography), Software Engineering Intern June - September 2025

- Created a CI/CD automated workflow for FPGA engineers to help streamline synthesis, simulation, and automated testing
- Developed a virtual environment for engineers to standardize software and hardware production

Research on the BPE Algorithm, (Prof. Konstantin Makarychev), Researcher January 2025 – May 2026

- Analyzed algorithms for the tokenization step of LLMs such as ChatGPT
- Constructed proofs to improve upper and lower bounds of the Byte Pair Encoding (BPE) algorithm

NSF REU Research in Computational Geometry, (Prof. Csaba Toth), Researcher June - August 2023

- Developed spanner proofs which are used in computational geometry to solve proximity problems
- Focused on creating spanners of intersection graphs with fewest edges in a given hop distance
- Created upper and lower bound proofs for 2-hop spanners of wedges, unit squares, and unit cubes

8vdX, Software Engineering Intern June - August 2022

- Built dynamic charts and forms for displaying Key Performance Indicators for 8vdX, an India-based, Y Combinator funded fintech company, that matched start-ups with venture debt investors

Technion Department of Computer Science, (Prof. Gershon Elber) Researcher June - August 2021

- Created customizable semiregular geometric tiles used to make 3D printed objects
- Published as 2nd author on paper titled "[Shell-lattice construction based on regular and semi-regular tiling via functional composition](#)"

PROJECTS

Project Hybrot Fall 2023 - Spring 2024

- Created software to stimulate neurons in a closed loop to observe how they learn information, set up and record experiments, and analyze resulting data

Terminal Chess Winter 2023 - Summer 2023

- Programmed chess played in terminal with board and piece customization, game variants, and bot levels

SKILLS & INTERESTS

Languages: C, C++, C#, JavaScript, Python, Java, HTML, CSS, x86 ASM

Software Programs/Tools: Shell Scripting, Unity, GitHub, Bitbucket, ReactJS

Interests: Puzzles, Chess, Rubik's Cubes, Premier League (Chelsea), Fantasy (TV series/Books), Baking